

```

import numpy as np

# Sigmoid function
def sigmoid(x):
    return 1 / (1 + np.exp(-x))

# Derivative of sigmoid
def sigmoid_derivative(x):
    return x * (1 - x)

# Input dataset (Example: XOR problem)
X = np.array([[0,0],
               [0,1],
               [1,0],
               [1,1]])

# Output dataset
y = np.array([[0],
               [1],
               [1],
               [0]])

# Initialize weights and biases
np.random.seed(1)
input_neurons = 2
hidden_neurons = 2
output_neurons = 1

# Weights
W1 = np.random.uniform(size=(input_neurons, hidden_neurons))
W2 = np.random.uniform(size=(hidden_neurons, output_neurons))

# Biases
b1 = np.random.uniform(size=(1, hidden_neurons))
b2 = np.random.uniform(size=(1, output_neurons))

learning_rate = 0.1

# Training
for epoch in range(10000):

    # Forward Propagation
    hidden_input = np.dot(X, W1) + b1
    hidden_output = sigmoid(hidden_input)

    final_input = np.dot(hidden_output, W2) + b2
    final_output = sigmoid(final_input)

    # Error
    error = y - final_output

```

```

# Back Propagation
d_output = error * sigmoid_derivative(final_output)

error_hidden = d_output.dot(W2.T)
d_hidden = error_hidden * sigmoid_derivative(hidden_output)

# Update weights and biases
W2 += hidden_output.T.dot(d_output) * learning_rate
W1 += X.T.dot(d_hidden) * learning_rate

b2 += np.sum(d_output, axis=0, keepdims=True) * learning_rate
b1 += np.sum(d_hidden, axis=0, keepdims=True) * learning_rate

# Output
print("Final Output after Training:")
print(final_output)

Final Output after Training:
[[0.07304485]
 [0.93084242]
 [0.93122512]
 [0.07564609]]

```